34. Hypercomplex Quantities and Representation Theory

Continuation and completion: Chapter III, §§15–19, and Chapter IV, §§20–26.

Chapter III. Module and representation theory

§ 15. Representations and representation modules

Let $\mathfrak{o}$ be a ring and let K be a ring with identity element. In later applications K will always be a field.

A representation of degree n of $\mathfrak{o}$ in K is a homomorphism

$$\mathfrak{o} \sim \mathfrak{D},$$

where $\mathfrak{D}$ is a ring of matrices of degree n over K.

By a *representation module* of $\mathfrak{o}$ with respect to K one understands a double module $\mathfrak{M}$ which is a left $\mathfrak{o}$-module and a right K-module,

$$\mathfrak{o}\mathfrak{M} \subseteq \mathfrak{M}, \qquad \mathfrak{M}K \subseteq \mathfrak{M},$$

which is also a direct sum of finitely many one-membered K-modules,

$$\mathfrak{M} = x_1 K + \cdots + x_n K,$$

and in which the identity element of K is the identity operator.

Every representation module leads to a representation. Let $c \in \mathfrak{o}$ and

$$cx_k = \sum_i x_i \gamma_{ik},$$

or, in matrix form,

$$c(x_1, \dots, x_n) = (x_1, \dots, x_n)C, \qquad C = (\gamma_{ik}).$$

Then the matrices C form a representation of c, for

$$(b + c)x_k = \sum_i x_i(\beta_{ik} + \gamma_{ik}),$$

and

$$\begin{aligned} bcx_k = b\sum_j x_j\gamma_{jk} &= \sum_j bx_j\gamma_{jk} = \sum_j \sum_i x_i\beta_{ij}\gamma_{jk} \\ &= \sum_i x_i\Big(\sum_j \beta_{ij}\gamma_{jk}\Big), \end{aligned}$$

or

$$bc(x_1, \dots, x_n) = (x_1, \dots, x_n)BC.$$

Conversely, every representation $\mathfrak{D}$ of $\mathfrak{o}$ belongs to a representation module, indeed to a definite basis of that module. Namely, let $\mathfrak{M}$ be the totality of the formal linear forms

$$y = \sum_i x_i\alpha_i, \qquad \alpha_i \in K.$$

Then $\mathfrak{M}$ is a right K-module. Define further, if the element c is assigned the matrix $C = (\gamma_{ik})$,

$$cx_k = \sum_i x_i\gamma_{ik}, \qquad c\Big(\sum_k x_k\alpha_k\Big) = \sum_i x_i \sum_k \gamma_{ik}\alpha_k.$$

From the homomorphism relations

$$c + d \sim C + D, \qquad cd \sim CD$$

it follows that

$$(c + d)x_i = cx_i + dx_i, \qquad cdx_i = c(dx_i),$$

and the same is true for sums $\sum_i x_i\alpha_i$. The remaining double-module laws

$$c(y+z) = cy + cz, \qquad c(y\alpha) = (cy)\alpha$$

are trivial. Thus $\mathfrak{M}$ is a representation module and, by construction, belongs exactly to the representation $\mathfrak{o} \sim \mathfrak{D}$.

Let $(y_1, \ldots, y_n)$ be another basis for the same representation module:

$$(y_1, \ldots, y_n) = (x_1, \ldots, x_n)P, \qquad (x_1, \ldots, x_n) = (y_1, \ldots, y_n)P^{-1}.$$

If $b \sim B$ with respect to the x-basis, then with respect to the y-basis

$$b(y_1, \ldots, y_n) = b(x_1, \ldots, x_n)P = (x_1, \ldots, x_n)BP = (y_1, \ldots, y_n)P^{-1}BP.$$

One calls the representations $b \sim B$ and $b \sim P^{-1}BP$ *equivalent* and counts them in the same representation class. Since every regular matrix P transforms one basis of $\mathfrak{M}$ into another, we have proved:

Every representation module leads uniquely to one definite representation class.[15a)]

15) Representation modules with respect to commutative fields occur first in W. Krull, *Theorie und Anwendung der verallgemeinerten Abelschen Gruppen*, Sitzungsberichte der Heidelberger Akademie, 1926, 1st treatise.

15a) Added in proof (14 April 1929). As B. L. van der Waerden informed me, one can obtain an invariant connection independent of the special choice of basis by separating the notions "linear transformation" and "matrix." A linear transformation is a homomorphism of two modules of linear forms; a matrix is the expression, or representation, of this homomorphism for a particular choice of basis. I. Every representation module assigns to the multiplier domain $\mathfrak{o}$ a unique homomorphic system of linear transformations of the module into itself. Indeed, by the end of §2 the left multipliers of a double module $\mathfrak{M}$ generate operator homomorphisms of $\mathfrak{M}$ into itself with respect to the right operators (here multiplication by K), and the assignment is homomorphic. II. Every system of linear transformations of a linear-form module into itself that is homomorphic to $\mathfrak{o}$ leads to a representation module.

It is clear that two operator-isomorphic representation modules correspond to the same representation class. Conversely, if two representation modules $(x_1, \ldots, x_n)$ and $(\bar{x}_1, \ldots, \bar{x}_n)$ generate the same representation, then the assignment

$$x_i \mapsto \bar{x}_i, \qquad \sum_i x_i\alpha_i \mapsto \sum_i \bar{x}_i\alpha_i$$

gives an isomorphism. For the multiplication rule is completely determined by the matrices C.

Special case: if two bases of $\mathfrak{M}$ generate the same representation, they are carried into one another by an operator isomorphism of $\mathfrak{M}$ onto itself.

Equivalently, if $C = PCP^{-1}$ for all C and one fixed P, then the assignment

$$y_i \mapsto x_i, \qquad y_i = \sum_k x_k\pi_{ik}$$

is an isomorphism of $\mathfrak{M}$ onto itself. Conversely, every isomorphism $y_i \mapsto x_i$ of the double module $\mathfrak{M}$ onto itself is mediated by a matrix P which commutes with all C.

One may also extend the notion of representation to rings which are still commutatively linked with a commutative multiplier domain P (§8). In that case one takes only such representation rings K as contain P in their center, and one demands of the representation not merely that it be a ring homomorphism $\mathfrak{o} \sim \mathfrak{D}$, but that it be an operator homomorphism: from $a \sim A$ there should follow

$$a\rho \sim A\rho \qquad (\rho \in P).$$

For the representation modules this requirement means the additional rule

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

The module and the ring are then said to be *commutatively connected with* P.

§ 16. Reducible representations

If $\mathfrak{U}$ is a submodule of the representation module $\mathfrak{M}$, and if it is possible to choose a basis for $\mathfrak{M}$ consisting of a basis $z_1, \ldots, z_t$ of $\mathfrak{U}$, supplemented by $y_1, \ldots, y_r$, thus

$$\mathfrak{M} = y_1K + \cdots + y_rK + z_1K + \cdots + z_tK, \tag{2a}$$

then the representations have the form

$$C = \begin{pmatrix} R & 0 \\ S & T \end{pmatrix},$$

where the matrices T by themselves form a representation of degree t mediated by $\mathfrak{U}$, and the matrices R form a representation of degree r mediated by $\mathfrak{M}/\mathfrak{U}$.

Indeed,

$$(cy_1, \dots, cy_r, cz_1, \dots, cz_t) = (y_1, \dots, y_r, z_1, \dots, z_t)C.$$

Since the cz_i, as elements of $\mathfrak{U}$, are expressible through the z_i alone, the upper-right block of C is zero. If the other parts of C are named R, S, T in the order shown above, then in particular

$$(cz_1, \dots, cz_t) = (z_1, \dots, z_t)T;$$

hence the T form a representation mediated by $\mathfrak{U}$. Further,

$$(cy_1, \dots, cy_r) \equiv (y_1, \dots, y_r)R \pmod{\mathfrak{U}},$$

while the y_i form a basis linearly independent modulo $\mathfrak{U}$. Thus the R form a representation generated by $\mathfrak{M}/\mathfrak{U}$.

Conversely, if a "reducible" representation

$$C = \begin{pmatrix} R & 0 \\ S & T \end{pmatrix}$$

is given, with R and T square matrices, then in the associated representation module the products of each c with the last t basis elements $z_1, \dots, z_t$ are expressed through those elements alone; hence $\mathfrak{U} = (z_1, \dots, z_t)$ is a submodule.

Corollary. Let

$$\mathfrak{M} \supset \mathfrak{U}_1 \supset \mathfrak{U}_2 \supset \cdots \supset \mathfrak{U}_e = 0$$

be a composition series for $\mathfrak{M}$, and suppose that each $\mathfrak{U}_i$ is a direct summand of the preceding one as a K-module. Then one can choose a basis

$$z_{11}, \dots, z_{1r_1}, \quad z_{21}, \dots, z_{2r_2}, \quad \dots \quad z_{e1}, \dots, z_{er_e}$$

such that the z_{ik} with $i > \nu$ form a basis of $\mathfrak{U}_\nu$. The representations mediated by $\mathfrak{M}$ then have the form

$$C = \begin{pmatrix} R_{11} & 0 & \cdots & 0 \\ R_{12} & R_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ R_{1e} & R_{2e} & \cdots & R_{ee} \end{pmatrix}, \tag{2}$$

and the $R_{\nu\nu}$ form representations mediated by the composition factors $\mathfrak{U}_{\nu-1}/\mathfrak{U}_\nu$.

The individual representations $R_{\nu\nu}$ are irreducible, since the composition factors $\mathfrak{U}_{\nu-1}/\mathfrak{U}_\nu$ are simple.

If K is assumed to be a field, so that every submodule is a direct summand, then conversely every representation reduced as far as possible in the form (2) leads to a composition series. By the Jordan-Hölder theorem the composition factors are uniquely determined up to operator isomorphism. Hence the $R_{\nu\nu}$ are uniquely determined up to equivalent representations and up to their order.[16)]

16) In other words: if $\mathfrak{U}$ is a direct summand as a K-module. If K is a field, this hypothesis is always fulfilled.

If $\mathfrak{M} = \mathfrak{A} + \mathfrak{B}$ is the direct sum of two representation modules $\mathfrak{A} = (y_1, \dots, y_r)$ and $\mathfrak{B} = (z_1, \dots, z_s)$, then the representation mediated by the basis $(y_1, \dots, y_r, z_1, \dots, z_s)$ has the form

$$C = \begin{pmatrix} R & 0 \\ 0 & S \end{pmatrix},$$

where R denotes the representation of the element c mediated by $\mathfrak{A}$, and S the one mediated by $\mathfrak{B}$; conversely as well. If one first writes $\mathfrak{M}$ as a direct sum of directly indecomposable components and draws composition series in these as above, then for a suitable basis the representation has a block form whose large boxes correspond to the directly indecomposable components and whose diagonal subblocks correspond to the composition factors.

If again K is a field, every representation reduced as far as possible in this way conversely leads to a representation of the module by directly indecomposable summands and composition factors within them. By the theorem of Krull and Otto Schmidt, the classes of directly indecomposable representation components are uniquely determined up to order.

If the module $\mathfrak{M}$ is completely reducible, all boxes in the block form consist of a single irreducible representation, and one calls the representation completely reducible.

§ 17. Direct sum decompositions of representation modules for rings with identity element

Let $\mathfrak{M}$ be an $\mathfrak{o}$-module, with $\mathfrak{o}$ a ring with identity. Then

$$\mathfrak{M} = \mathfrak{M}_e + \mathfrak{M}_0,$$

where the identity is the identity operator on $\mathfrak{M}_e$ and the zero operator on $\mathfrak{M}_0$. If $\mathfrak{M}$ is also a right K-module, then $\mathfrak{M}_e$ and $\mathfrak{M}_0$ are also right K-modules.

Proof. Every $m \in \mathfrak{M}$ can be written as

$$m = em + (m - em).$$

Let the set of the em be $\mathfrak{M}_e$, and the set of the $m - em$ be $\mathfrak{M}_0$. These are plainly additive groups, and right K-modules if $\mathfrak{M}$ is one. Moreover

$$r(em) = erm,$$

so $\mathfrak{M}_e$ is also an $\mathfrak{o}$-module; and

$$r(m - em) = rm - rem = rm - rm,$$

so $\mathfrak{M}_0$ is an $\mathfrak{o}$-module. For the em, e is identity operator; for the $m - em$, it is zero operator. Thus only zero belongs to both $\mathfrak{M}_e$ and $\mathfrak{M}_0$, and the sum $\mathfrak{M}_e + \mathfrak{M}_0$ is direct.

For representation modules, $\mathfrak{M}_0$ gives the trivial representation, in which every element is assigned zero. Splitting off the summand $\mathfrak{M}_0$, there remains the representation mediated by $\mathfrak{M}_e$, in which the identity element is assigned the identity matrix. We can and will therefore restrict ourselves henceforth to such representations.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct sum of two-sided ideals,

$$e = e_1 + \cdots + e_s$$

the corresponding decomposition of e, and let $\mathfrak{M}$ be an $\mathfrak{o}$-module on which e is identity operator. Then

$$\mathfrak{M} = \mathfrak{a}_1\mathfrak{M} + \cdots + \mathfrak{a}_s\mathfrak{M}$$

directly.

Indeed, plainly $\mathfrak{M} = \mathfrak{o}\mathfrak{M} = (\mathfrak{a}_1\mathfrak{M}, \ldots, \mathfrak{a}_s\mathfrak{M})$. Put $\mathfrak{b}_i = \mathfrak{a}_1 + \cdots + \widehat{\mathfrak{a}_i} + \cdots + \mathfrak{a}_s$. Then correspondingly $\mathfrak{b}_i$ is a two-sided ideal and

$$\mathfrak{M} = (\mathfrak{a}_i\mathfrak{M}, \mathfrak{b}_i\mathfrak{M}),$$

and this sum is direct, since e_i is identity operator on $\mathfrak{a}_i\mathfrak{M}$ and zero operator on $\mathfrak{b}_i\mathfrak{M}$.

On the basis of this theorem we shall usually restrict ourselves to representations of two-sided indecomposable rings.

§ 18. Module and representation theory of completely reducible rings

Let $\mathfrak{o}$ be completely reducible and two-sided simple, and let $\mathfrak{M}$ be a finite $\mathfrak{o}$-module for which the identity element of $\mathfrak{o}$ is identity operator. Then $\mathfrak{M}$ is completely reducible, and its simple constituents are operator-isomorphic to the simple left ideals $\mathfrak{l}_i$.

Proof. Let

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n, \qquad \mathfrak{M} = (\mathfrak{o}m_1, \ldots, \mathfrak{o}m_k),$$

and therefore

$$\mathfrak{M} = (\ldots, \mathfrak{l}_i m_k, \ldots). \tag{4}$$

The non-zero $\mathfrak{l}_i m_k$ are isomorphic to $\mathfrak{l}_i$, since the assignment $a \mapsto am_k$ is an operator isomorphism. Hence the $\mathfrak{l}_i m_k$ are simple. Omitting from (4) those $\mathfrak{l}_i m_k$ already contained in the sum of the preceding ones, the sum becomes direct.

Consequently, if in addition $\mathfrak{M}$ is directly indecomposable, then $\mathfrak{M}$ is simple and isomorphic to some $\mathfrak{l}_i$.

Let Δ be the automorphism field of this simple $\mathfrak{M}$, and let Λ be that of $\mathfrak{l}_i$. Then, by the operator isomorphism of $\mathfrak{M}$ and $\mathfrak{l}_i$, we have the ring isomorphism $\Delta \simeq \Lambda$. By §1 one can regard $\mathfrak{M}$ and $\mathfrak{l}_i$ as double modules with Δ and Λ, respectively, as right domains; they are then also representation modules. The representations mediated by them therefore pass into one another under the ring isomorphism $\Delta \simeq \Lambda$. Thus for the study of this representation class we may restrict ourselves to the representation mediated by $\mathfrak{l}_i$ in its automorphism field.

Specifically, put a basis of matrix units for $\mathfrak{l}_1$ in the form

$$\mathfrak{l}_1 = (c_{11}, \ldots, c_{n1})$$

and realize the automorphism field Λ of the earlier §14 by the opposite field of $\mathfrak{o}$; in the earlier considerations right ideals are to be replaced by left ideals and automorphisms written on the right. If

$$c = \sum_{i,k} c_{ik} \alpha_{ik}$$

is an element of $\mathfrak{o}$, then

$$(cc_{11}, \ldots, cc_{n1}) = (c_{11}, \ldots, c_{n1})(\alpha_{ik}).$$

Thus the representation mediated by $\mathfrak{l}_1$ is given by the matrix (α_{ik}). For $\mathfrak{l}_\nu = (c_{1\nu}, \ldots, c_{n\nu})$ one obtains exactly the same representation.

If $\mathfrak{o}$ is a direct sum of two-sided simple components, then these components are represented in turn isomorphically, while the remaining components act by zero.

§ 19. The simple composition factors in modules and representation modules

Let $\mathfrak{o}$ be a ring with identity, and let $\mathfrak{c}$ be its radical. In every simple module and every simple representation module over $\mathfrak{o}$, the elements of $\mathfrak{c}$ have annihilator effect; hence one obtains a module or representation module for $\mathfrak{o}/\mathfrak{c}$.

Indeed, $\mathfrak{c}\mathfrak{M}$ is again a module (and, in the representation case, also a right K-module). Since $\mathfrak{M}$ is simple, either $\mathfrak{c}\mathfrak{M} = \mathfrak{M}$ or $\mathfrak{c}\mathfrak{M} = 0$. In the former case $\mathfrak{c}^\rho\mathfrak{M} = \mathfrak{M}$ for every exponent ρ, contradicting nilpotence of $\mathfrak{c}$. Thus $\mathfrak{c}\mathfrak{M} = 0$.

It follows that the simple composition factors of arbitrary modules and representation modules are also generated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

For representation modules commutatively connected with P, this means in particular that, on reducing an arbitrary representation by means of a composition series as in §16, only the diagonal representations mediated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$ occur, apart from zeros.

No finiteness condition on $\mathfrak{o}$ beyond the hypotheses already placed on the representation over P is needed for these facts. Every representation is an isomorphic representation of its absolute multiplier domain. This absolute multiplier domain, as inverse image of a finite matrix ring, is itself a finite-rank P-module. Since the permitted ideals are by definition P-modules, the maximal and minimal conditions hold for it.

If P is a commutative field, this absolute multiplier domain is a hypercomplex system over P. This assumption is automatically fulfilled whenever non-zero representations occur on the diagonal: then P is isomorphic to a subfield of the automorphism field of a simple left ideal, and, in realization by the subfield K of $\mathfrak{o}/\mathfrak{c}$, the commutative connection forces it into the center of K.

Chapter IV. Representations of groups and hypercomplex systems

§ 20. Inclusion of hypercomplex systems

We now consider hypercomplex systems $\mathfrak{o}$ with respect to a commutative field P. By convention, only P-modules are counted as ideals in $\mathfrak{o}$. Thus the maximal and minimal conditions for left and right ideals are fulfilled, and the entire ring theory of Chapter II applies.

In what follows, by a representation of the hypercomplex system $\mathfrak{o}$ we always mean one in the field P, and more precisely one satisfying: from $a \sim A$ follows

$$a\rho \sim A\rho \qquad (\rho \in P),$$

i.e. module homomorphy with respect to P. For representation modules this reads

$$a\rho \cdot m = a(m\rho).$$

The results of §19 therefore apply. All irreducible representations are mediated by simple left ideals of

$$\mathfrak{o}_0 = \mathfrak{o}/\mathfrak{c},$$

where $\mathfrak{c}$ is the radical. Thus there are as many inequivalent irreducible representations as there are two-sided simple summands $\mathfrak{a}$ in the decomposition

$$\mathfrak{o}_0 = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s.$$

To determine explicitly the representation defined by such a left ideal, one first passes from the elements of $\mathfrak{o}$ to their residue classes in $\mathfrak{o}_0$. Multiplication by an element of P is an isomorphism for all ideals of $\mathfrak{o}_0$; hence P is isomorphic to a subfield $e^{(\nu)}P$ of the automorphism field K_ν of every simple left ideal. The representation mediated by $\mathfrak{l}_\nu$ over P is found by first considering the representation over this subfield $e^{(\nu)}P$ and then transferring it to P by the isomorphism $e^{(\nu)}P \simeq P$. The field K_ν has finite rank over P, so this is a representation in a finite-degree subfield of the automorphism field of $\mathfrak{l}_\nu$.

Let

$$c = c_1 + \cdots + c_s$$

be the decomposition of an element of $\mathfrak{o}_0$ into its two-sided components. For the representation mediated by $\mathfrak{l}_\nu$, only the matrix of c_ν is considered, since the other c_i annihilate $\mathfrak{l}_\nu$ and are represented by zero matrices. If one writes c_ν in matrix units as

$$c_\nu = \sum c_{ik}^{(\nu)} \alpha_{ik}^{(\nu)},$$

then every $\alpha_{ik}^{(\nu)}$ is to be replaced by the matrix that represents it in the representation of K_ν over P.

Since K_ν has finite rank over P, every element x of K_ν satisfies an equation $f(x) = 0$ with coefficients in $e^{(\nu)}P$, because its powers are linearly dependent. If P is algebraically closed, the equation splits into linear factors; since K_ν is a field, x is already a root of one of the linear factors, and hence lies in $e^{(\nu)}P$. Thus $K_\nu = e^{(\nu)}P$. In this case the matrices $(\alpha_{ik}^{(\nu)})$ themselves are the irreducible representations.

Together with the end of §19 this gives Burnside's theorem:

Let P be algebraically closed and commutatively connected with a ring $\mathfrak{o}$. Then every irreducible representation of degree n over P contains exactly n^2 linearly independent matrices.

From this follows at once the customary formulation: a system of matrices of degree n over P, closed under multiplication, is irreducible if and only if no non-trivial homogeneous linear relation

$$\sum \alpha_{ik} x_{ik} = 0$$

with coefficients in P exists that is satisfied by the entries x_{ik} of all the matrices. Indeed, adjoining all linear combinations, one obtains a ring and a P-module and can regard this ring as its own representation.

For the absolute multiplier domain is isomorphic to a two-sided simple completely reducible ring with identity, hence to a full matrix ring over the automorphism field of its left ideals. The algebraic closedness of P makes that automorphism field equal to P. The irreducible representation is generated by a simple left ideal. If this ideal has rank m, and the representation has degree n, then $m = n$, and in the matrix ring there are exactly n^2 linearly independent elements.

Similarly, the generalized Burnside theorem follows. Under the same hypotheses, if $\mathfrak{D}$ is a completely reducible representation decomposed into inequivalent components of degrees

$$n_1, n_2, \ldots, n_s,$$

then $\mathfrak{D}$ has rank

$$n_1^2 + n_2^2 + \cdots + n_s^2$$

with respect to P.

If $\mathfrak{o}$ is a hypercomplex system without radical over an arbitrary field, then $\mathfrak{o}$ is completely reducible and has an identity element. From §§17 and 18 it follows that every representation of $\mathfrak{o}$ is completely reducible.

Conversely, every completely reducible representation of a ring $\mathfrak{o}$ commutatively connected with P has as absolute multiplier domain a hypercomplex system without radical. For the absolute multiplier domain is isomorphic to a ring and P-module of matrices over P, hence is a hypercomplex system. The elements of its radical are represented by zero in every irreducible constituent, hence in the whole representation; therefore the radical is zero.

The *regular representation* of a hypercomplex system $\mathfrak{o}$ is the representation mediated by the unit ideal $\mathfrak{o}$ itself. For a group ring, one chooses specifically the group elements as basis. Since all left ideals of $\mathfrak{o}/\mathfrak{c}$ occur as composition factors in $\mathfrak{o}$, all irreducible representations already occur on the diagonal of the regular representation after the regular representation is reduced by means of a composition series, as in §16.

A representation is known when the representing matrices for the basis elements $u_1, \ldots, u_h$ are known. If the system is a group ring, so that the u_i are group elements, then every homomorphic matrix representation of the group is also a representation of the group ring. Thus the problem of representing a group is a special case of representing hypercomplex systems.

§ 21. Extension of the ground field. Representations of the center

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system, and let Ω be an extension field of P. One can form

$$\mathfrak{o}\Omega = a_1 \Omega + \cdots + a_h \Omega$$

with the old multiplication rules for the basis elements.[18a] Then $\mathfrak{o}\Omega$ is again hypercomplex with respect to Ω.

Every representation of $\mathfrak{o}$ leads to a representation of $\mathfrak{o}\Omega$, since the representation of all elements is known as soon as the representations of the basis elements a_i are known.[19]

18a) More precisely: one introduces new symbols a_i with the old multiplication rules; $a_1\Omega + \cdots + a_h\Omega$ then contains a subring isomorphic to $\mathfrak{o}$, so that the new symbols may afterwards be identified with the old a_i.

19) Of course, here again only representations which are P-module homomorphisms are considered: from $a \sim A$ should follow $a\rho \sim A\rho$ for $\rho \in P$, and correspondingly for Ω.

An ideal or representation module, as well as the corresponding representation, is called *absolutely irreducible* if it remains irreducible after passage to the algebraically closed field.

If $\mathfrak{o}\Omega$ is without radical, then so is $\mathfrak{o}$; for a nilpotent ideal $\mathfrak{c}$ in $\mathfrak{o}$ would lead to an extended ideal $\mathfrak{c}\Omega$ in $\mathfrak{o}\Omega$. The converse is not true, as will be seen below.

The center of $\mathfrak{o}\Omega$ is equal to the extended center $\mathfrak{Z}\Omega$. If $\mathfrak{o}$ is completely reducible, then $\mathfrak{Z}$ is a direct sum of fields:

$$\mathfrak{Z} = \mathfrak{Z}_1 + \cdots + \mathfrak{Z}_s.$$

If $\mathfrak{o}\Omega$ remains completely reducible upon passage to Ω (that is, if no nilpotent ideal is added), then the new center decomposes as

$$\mathfrak{Z}\Omega = \mathfrak{Z}_1\Omega + \cdots + \mathfrak{Z}_s\Omega$$

again into a direct sum of fields. If Ω is algebraically closed, these fields have rank 1 over Ω and are isomorphic to Ω.

For the representations of the center $\mathfrak{Z}$ the following theorems hold.

Every irreducible representation of a commutative hypercomplex system $\mathfrak{Z}$ in the algebraically closed field Ω is of degree 1; equivalently, the irreducible representations are identical with the homomorphisms $\mathfrak{Z} \to \Omega$.

Proof. Every irreducible representation of $\mathfrak{Z}$ yields one of $\mathfrak{Z}\Omega$, hence also one of $\mathfrak{Z}\Omega/\mathfrak{c}$, where $\mathfrak{c}$ is the radical of $\mathfrak{Z}\Omega$. The quotient $\mathfrak{Z}\Omega/\mathfrak{c}$ is a commutative system without radical, hence by the end of §14 a direct sum of fields. These are of degree 1 because Ω is algebraically closed, and they generate all irreducible representations (§20).

At the same time, since equivalent representations of degree 1 necessarily become equal, the number of distinct homomorphisms $\mathfrak{Z} \to \Omega$ is equal to the rank of $\mathfrak{Z}\Omega/\mathfrak{c}$.

In the special case that $\mathfrak{Z}$ is a field over P, this gives the theorem: $\mathfrak{Z}$ is completely reducible, i.e. without radical, if and only if $\mathfrak{Z}$ is an extension of the first kind of P. For a field the homomorphisms are isomorphisms. Their number is the rank of $\mathfrak{Z}\Omega/\mathfrak{c}$, and it is equal to the field degree precisely when $\mathfrak{c} = 0$.

If

$$\mathfrak{Z} = \mathfrak{Z}_1 + \cdots + \mathfrak{Z}_s$$

is completely reducible, then in each representation the separate fields $\mathfrak{Z}_i$ are also represented homomorphically; in an irreducible representation one of these fields is mapped isomorphically and the others by zero.

For systems without radical, the application of these facts to hypercomplex systems gives first:

If $\mathfrak{o}\Omega$, and therefore also $\mathfrak{o}$, is a system without radical, then in every absolutely irreducible representation of $\mathfrak{o}$ the central elements z are represented by diagonal matrices $E\xi$, and $z \mapsto \xi$ is a representation of degree 1 of $\mathfrak{Z}$ in Ω. The resulting correspondence between the absolutely irreducible representation classes of $\mathfrak{o}$ and those of $\mathfrak{Z}$ is one-to-one. Thus the number of these representation classes is equal to the rank of the center.[20)]

20) Corresponding theorems hold not only for representations in the algebraically closed field Ω, but also for representations in the individual automorphism fields; this will be carried out elsewhere.

Indeed, the decomposition into two-sided indecomposable ideals

$$\mathfrak{o}\Omega = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

corresponds one-to-one to a decomposition of the center into fields

$$\mathfrak{Z}\Omega = \mathfrak{Z}_1 + \cdots + \mathfrak{Z}_s,$$

where $\mathfrak{Z}_i$ is the center of $\mathfrak{a}_i$. In an irreducible representation of $\mathfrak{o}$, one $\mathfrak{a}_i$ is mapped isomorphically and the others by zero. The representation class is uniquely determined by $\mathfrak{a}_i$. The component $\mathfrak{a}_i$ is represented by a full matrix ring, and the center of such a full matrix ring consists of diagonal matrices $E\xi$. The assignment $z \mapsto \xi$ is a homomorphism of $\mathfrak{Z}$ in which one $\mathfrak{Z}_i$ is mapped isomorphically and the others by zero. This proves the assertion.

For the question when a radical-free $\mathfrak{o}$ gives rise to a radical-free $\mathfrak{o}\Omega$, it follows that if $\mathfrak{Z}$ has a component $\mathfrak{Z}_i$ which is an extension of the second kind of P, then $\mathfrak{o}\Omega$ certainly has a radical.[21)] For $\mathfrak{Z}_i\Omega$ already has one in that case.

21) If $\mathfrak{Z}$ has only components of the first kind, then $\mathfrak{o}\Omega$ is without radical, as will likewise be proved later. For characteristic zero I. Schur already proved the equivalent theorem that irreducible representations over P remain completely reducible under every extension of the coefficient domain; see I. Schur, Beiträge zur Theorie der Gruppen linearer Substitutionen, *Trans. Amer. Math. Soc.* 15 (1909), p. 159.

§ 22. Application to abelian groups

Let

$$G = G_1 \times G_2 \times \cdots \times G_r$$

be a finite abelian group, decomposed into cyclic groups G_i of orders h_i. Its elements are

$$a = a_1^{\alpha_1} \cdots a_r^{\alpha_r}.$$

Let P be a field whose characteristic does not divide the group order

$$h = h_1 h_2 \cdots h_r.$$

The group ring $\mathfrak{Z}$ consists of all sums

$$\sum_{\alpha_1,\ldots,\alpha_r} A_{\alpha_1\ldots\alpha_r} a_1^{\alpha_1} \cdots a_r^{\alpha_r} \qquad (A \in P)$$

and is a homomorphic image of the polynomial ring $P[x_1, \ldots, x_r]$ through $x_i \mapsto a_i$. Thus

$$\mathfrak{Z} \simeq P[x_1, \ldots, x_r]/(x_1^{h_1} - 1, \ldots, x_r^{h_r} - 1).$$

In the extension field Ω the polynomials $x_i^{h_i} - 1$ split into distinct factors $x_i - \xi_i$, so the defining ideal is the product, or intersection, of distinct maximal ideals

$$(x_1 - \xi_1^{(\alpha_1)}, \ldots, x_r - \xi_r^{(\alpha_r)}),$$

one for each zero $(\xi_1^{(\alpha_1)}, \ldots, \xi_r^{(\alpha_r)})$. To this intersection corresponds a direct-sum decomposition

$$\mathfrak{Z}\Omega = \mathfrak{Z}_1 + \cdots + \mathfrak{Z}_h,$$

where each component is isomorphic to Ω and gives the representation

$$a_i \mapsto \xi_i^{(\alpha_i)}, \qquad \text{or more generally} \qquad a \mapsto \chi^{(\alpha)}(a).$$

These representations are the characters. Their number is $h_1 \cdots h_r = h$, as the general theory requires.

§ 23. Determinant of a hypercomplex system

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. Adjoin to P the h indeterminates $x_1, \ldots, x_h$ and form

$$\mathfrak{o}^* = a_1 P(x) + \cdots + a_h P(x),$$

where the old multiplication rules are used for the basis elements a_i. The x_i are to commute with the a_i, thereby fixing the rules of calculation in $\mathfrak{o}^*$.

In $\mathfrak{o}^*$ lies the "general element" of $\mathfrak{o}$,

$$w = a_1 x_1 + \cdots + a_h x_h.$$

If in a representation $a_i \mapsto A_i$, then w is assigned

$$W = A_1 x_1 + \cdots + A_h x_h.$$

The matrix W is called the system matrix belonging to the representation; if the a_i form a group and $\mathfrak{o}$ is the group ring, it is the group matrix. In the regular representation one has the regular system matrix.

The entries of W are linear forms in the x_i. The system determinant $|W|$ is therefore of degree n for a representation of degree n. In particular, the regular system determinant has degree h.

The system determinant is unchanged under passage to equivalent representations, since

$$|PWP^{-1}| = |P|\,|W|\,|P^{-1}| = |W|.$$

When one passes from the basis $(a_1, \ldots, a_h)$ to a new basis $(b_1, \ldots, b_h)$ and from $w = \sum a_i x_i$ to $w = \sum b_j y_j$, the new elements of W, and hence also the new determinant, are obtained from the old ones by a regular substitution of variables

$$x_i = \sum_j \rho_{ij} y_j.$$

If a composition series of the representation module is given, then for a suitable choice of basis the matrix W has lower block-triangular form,

$$W = \begin{pmatrix} W_1 & 0 & \cdots & 0 \\ * & W_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & W_s \end{pmatrix}.$$

Among the determinants $|W_i|$ of an arbitrary representation, no other factors occur than those occurring in the regular representation of $\mathfrak{o}$, or even of $\mathfrak{o}/\mathfrak{c}$, where $\mathfrak{c}$ is the radical.

If P is algebraically closed, then the determinant $|W_i|$ belonging to an irreducible representation is a prime function of the x_i, and inequivalent representations give different prime factors.

Proof. Since all irreducible representations of $\mathfrak{o}$ are also representations of $\mathfrak{o}/\mathfrak{c}$, we may restrict ourselves to the ring without radical $\mathfrak{o}/\mathfrak{c}$. In it take as basis the matrix units $c_{ik}^{(\nu)}$; the general element is then

$$w = \sum_{\nu,i,k} c_{ik}^{(\nu)} x_{ik}^{(\nu)}.$$

The matrices of the irreducible representations are

$$W_\nu = (x_{ik}^{(\nu)}).$$

The functions $|W_\nu| = |x_{ik}^{(\nu)}|$ are known to be irreducible and are plainly distinct.

To compute the $|W_i|$, one can factor the regular system matrix of $\mathfrak{o}/\mathfrak{c}$. Each prime factor occurs as often as the degree of the corresponding irreducible representation, since the corresponding ideal occurs that many times in the composition series.

One can also start from the regular system matrix of $\mathfrak{o}$, but then each irreducible factor is obtained more often, namely as often as the corresponding simple ideal occurs as a composition factor among the left ideals.

In this regular representation $\mathfrak{o}$ has been regarded as a left ideal. If $\mathfrak{o}$ is regarded as a right ideal, a second regular system matrix appears, the "antistrophic matrix" of Frobenius; it contains the same irreducible factors, namely the system determinants of all irreducible representations, though possibly with different exponents.

The system determinant of a commutative system splits into linear factors, because all irreducible representations have degree 1. These linear factors are the irreducible representations themselves and, for abelian groups, give the characters. This fact was Dedekind's point of departure in his study of the group determinant of non-abelian groups.

§ 24. Traces and characters

If, in a representation $\mathfrak{o} \sim \mathfrak{D}$ of a hypercomplex system $\mathfrak{o}$, the element a is assigned the matrix A, put

$$\mathrm{Sp}_D(a) = \mathrm{Sp}\, A.$$

Traces are linear functions:

$$\mathrm{Sp}(c+d) = \mathrm{Sp}(c) + \mathrm{Sp}(d), \qquad \mathrm{Sp}(c\alpha) = \alpha\, \mathrm{Sp}(c).$$

Equivalent representations have the same traces. In a reducible representation the trace is the sum of the traces in the representations mediated by the composition factors. Indeed,

$$\mathrm{Sp}\begin{pmatrix} A_{11} & 0 & \cdots & 0 \\ * & A_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & A_{ss} \end{pmatrix} = \mathrm{Sp}(A_{11}) + \mathrm{Sp}(A_{22}) + \cdots + \mathrm{Sp}(A_{ss}).$$

The *principal trace* is the trace in the regular representation. The *reduced trace* is the sum of the traces in the distinct irreducible representations.

If c is an element of the maximal nilpotent ideal $\mathfrak{c}$, then $\mathrm{Sp}(c) = 0$ for every representation.

Proof. It suffices to consider representations through the simple left ideals of $\mathfrak{o}/\mathfrak{c}$, since in any other representation only these composition factors occur, and the trace is additively composed. But c annihilates every element of $\mathfrak{o}/\mathfrak{c}$; hence the zero matrix is assigned to every $c \in \mathfrak{c}$, so $\mathrm{Sp}(c) = 0$.

If $\mathfrak{o}$ is two-sided simple and P algebraically closed, hence a matrix ring

$$\mathfrak{o} = \sum c_{ik} P,$$

then in the irreducible representation of $\mathfrak{o}$ the element

$$a = \sum c_{ik}\alpha_{ik}$$

is assigned the matrix (α_{ik}), so that

$$\mathrm{Sp}_{\mathrm{red}}(a) = \sum_i \alpha_{ii}, \qquad \mathrm{Sp}_{\mathrm{pr}}(a) = n\ \mathrm{Sp}_{\mathrm{red}}(a) = n \sum_i \alpha_{ii}.$$

In particular,

$$\mathrm{Sp}_{\mathrm{red}}(c_{ik}) = 0 \quad (i \neq k), \qquad \mathrm{Sp}_{\mathrm{red}}(c_{ii}) = e,$$

whereas

$$\mathrm{Sp}_{\mathrm{pr}}(c_{ik}) = 0 \quad (i \neq k), \qquad \mathrm{Sp}_{\mathrm{pr}}(c_{ii}) = ne.$$

Passing from P to the algebraically closed field Ω does not change the principal trace, since it can be computed from the same basis. This is not true of the reduced trace.

The traces of elements a of the system without radical $\mathfrak{o}$ in the absolutely irreducible representations are called characters and are denoted by $\chi(a)$, or by $\chi^{(\nu)}(a)$ when the representation is to be specified.

In an irreducible representation of degree n_ν, the central elements are represented, by §21, as diagonal matrices $E\theta^{(\nu)}$, where $z \mapsto \theta^{(\nu)}(z)$ is a representation of degree 1 of the center, or a homomorphism of the center into Ω. Therefore the trace has the value $n_\nu\theta^{(\nu)}(z)$. Thus the homomorphisms θ of the center are linked with the characters χ by

$$\chi^{(\nu)}(z) = n_\nu \theta^{(\nu)}(z). \tag{1}$$

In the commutative case $n_\nu = 1$, and the characters themselves give the homomorphisms.

If Ω has characteristic zero, as will always be assumed in what follows, one can divide (1) by n_ν:

$$\theta^{(\nu)}(z) = \frac{\chi^{(\nu)}(z)}{n_\nu}.$$

The homomorphism property of the $\theta^{(\nu)}(z)$ is expressed by

$$\frac{\chi^{(\nu)}(z+z')}{n_\nu} = \frac{\chi^{(\nu)}(z)}{n_\nu} + \frac{\chi^{(\nu)}(z')}{n_\nu},$$

and

$$\frac{\chi^{(\nu)}(zz')}{n_\nu} = \frac{\chi^{(\nu)}(z)}{n_\nu}\,\frac{\chi^{(\nu)}(z')}{n_\nu}.$$

A representation class is already uniquely determined by the traces of the matrices alone. To know the traces of all matrices it is of course enough to know the traces of the basis elements in the given representation.

Proof. The representation module R is known when one knows how often each irreducible representation module $\mathfrak{M}_\nu$ occurs in it as a direct summand. If this number is μ_ν, then the trace of $e^{(\nu)}$ in the representation is $\mu_\nu n_\nu$, where n_ν is the degree of the irreducible representation. Hence

$$\mu_\nu = \frac{\mathrm{Sp}(e^{(\nu)})}{n_\nu}.$$

§ 25. Discriminants

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. The *discriminant matrix* is the matrix whose entries are the principal traces $\mathrm{Sp}(a_i a_j)$. The *reduced discriminant matrix* is the corresponding matrix for reduced traces, formed in the algebraically closed field. The determinant of the matrix is the discriminant, respectively the reduced discriminant.

The discriminant remains the same under extension of the ground field. When passing to another basis it is multiplied by the square of the transformation determinant.

Proof. Let

$$(b_1, \ldots, b_h) = (a_1, \ldots, a_h)P.$$

Then

$$(\mathrm{Sp}(a_i a_j)) = (\mathrm{Sp}(a_i b_j))P. \tag{1}$$

Likewise, if P^t denotes the transposed matrix,

$$(\mathrm{Sp}(a_i b_j)) = P^t(\mathrm{Sp}(b_i b_j)), \tag{1a}$$

and from (1) and (1a)

$$(\mathrm{Sp}(a_i a_j)) = P^t(\mathrm{Sp}(b_i b_j))P.$$

Passing to determinants gives the assertion.

The determinant is therefore determined only up to a square factor from P, but its vanishing or non-vanishing is an invariant property. It also follows from (1) that, up to a non-zero factor, the discriminant may be computed from two different bases, namely as $|\,\mathrm{Sp}(a_i b_j)|$.

The discriminant vanishes if $\mathfrak{o}$ possesses a nilpotent ideal $\mathfrak{c}$.

Proof. Choose basis elements for $\mathfrak{o}$ in the form

$$c_1, \ldots, c_r, d_1, \ldots, d_s,$$

where $c_1, \ldots, c_r$ form a basis of $\mathfrak{c}$. Then the discriminant matrix has the form

$$\begin{pmatrix} \mathrm{Sp}(c_i c_j) & \mathrm{Sp}(c_i d_j) \\ \mathrm{Sp}(d_i c_j) & \mathrm{Sp}(d_i d_j) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \mathrm{Sp}(d_i d_j) \end{pmatrix}$$

in the top block sense needed for the determinant; hence the determinant is zero. The same holds for the reduced discriminant.

Consequently, the discriminant also vanishes if a nilpotent ideal appears after passage to the algebraically closed field.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct two-sided sum, and let $M, M_1, \ldots, M_s$ be the discriminant matrices of the rings $\mathfrak{o}, \mathfrak{a}_1, \ldots, \mathfrak{a}_s$. Then M is block diagonal,

$$M = \begin{pmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_s \end{pmatrix},$$

so

$$|M| = |M_1|\,|M_2| \cdots |M_s|.$$

Indeed, choose a basis of $\mathfrak{o}$ formed from bases of $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$. If $a \in \mathfrak{a}_i$, then

$$\mathrm{Sp}_{\mathfrak{o}}(a) = \mathrm{Sp}_{\mathfrak{a}_i}(a),$$

where both times principal traces are meant, but in the rings $\mathfrak{o}$ and $\mathfrak{a}_i$ respectively. Further, $a_i a a_j = 0$ if the factors lie in different two-sided components; hence the mixed trace blocks are zero. The same argument applies to the reduced discriminant.

To form the discriminant of a matrix ring $\sum c_{ik} P$, choose two different bases: the c_{ik} once ordered by first indices and once by second indices. The product table is

$$\begin{array}{cc|ccc}
 & & \overbrace{c_{11}\ \cdots\ c_{1n}}^{r_1} & \overbrace{c_{21}\ \cdots\ c_{2n}}^{r_2} & \cdots \\
\hline
\left.\begin{matrix} c_{11} \\ \vdots \\ c_{n1} \end{matrix}\right\} & t_1 & (c_{ik}) & 0 & 0 \\
\left.\begin{matrix} c_{12} \\ \vdots \\ c_{n2} \end{matrix}\right\} & t_2 & 0 & (c_{ik}) & 0 \\
\vdots & & 0 & 0 & (c_{ik})
\end{array}$$

and therefore the discriminant is

$$D = \begin{vmatrix} \mathrm{Sp}(c_{11}) & 0 & \cdots & 0 \\ 0 & \mathrm{Sp}(c_{22}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathrm{Sp}(c_{nn}) \end{vmatrix}^{n} = \begin{cases} e, & \text{for reduced traces,} \\ n^{n^2} e, & \text{for principal traces.} \end{cases}$$

Equivalently,

$$D_{\text{red}} = e \quad \text{for reduced traces,} \qquad D = n^{n^2} e \quad \text{for principal traces.}$$

Therefore, if over the algebraically closed extension field $\mathfrak{o}$ decomposes into matrix rings of degrees $n_1, \ldots, n_s$, its discriminant is

$$D = n_1^{n_1^2} n_2^{n_2^2} \cdots n_s^{n_s^2} e,$$

and its reduced discriminant is

$$D_{\text{red}} = e.$$

Thus the reduced discriminant is always non-zero for systems without radical; the principal discriminant is non-zero exactly when no n_i is divisible by the characteristic of the field. Hence:

Non-vanishing of the reduced discriminant is necessary and sufficient for systems without radical after algebraic closure of the field P.

In characteristic zero, non-vanishing of the discriminant is necessary and sufficient for the non-appearance of a radical after algebraic closure of the field P.[22)]

22) For commutative systems compare E. Noether, *Diskriminantensatz für Ordnungen …*, J. reine angew. Math. 157 (1927), pp. 82–104, §§4–6. The method of proof there is the same, but more complicated in detail; the transition from one basis to another (§4, 4 there) is to be replaced by the one given here at the beginning of this paragraph, for the determinant occurring there vanishes.

§ 26. Placement of the group ring

Let $a_1, \ldots, a_h$ be the elements of a finite group, and form the group ring over a field whose characteristic does not divide h. Then the discriminant $D \neq 0$, and hence the group ring is a ring without radical.

Proof. First we show, with Sp henceforth meaning principal trace,

$$\mathrm{Sp}(e) = he, \qquad \mathrm{Sp}(a_i) = 0 \quad (a_i \neq e).$$

In the regular representation one has

$$e \mapsto \begin{pmatrix} e & 0 & \cdots & 0 \\ 0 & e & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e \end{pmatrix}, \qquad \mathrm{Sp}(e) = he.$$

For $a_i \neq e$ the products $a_i a_1, \ldots, a_i a_h$ are just a permutation of $a_1, \ldots, a_h$ with no fixed point; therefore the matrix by which these products are expressed in terms of $a_1, \ldots, a_h$ has only zeros in the main diagonal, and so $\mathrm{Sp}(a_i) = 0$.

Take now the bases $a_1, \ldots, a_h$ and $a_1^{-1}, \ldots, a_h^{-1}$. The matrix $\big(\mathrm{Sp}(a_i a_j^{-1})\big)$ is diagonal with diagonal entries he; hence

$$D = h^h e \neq 0.$$

By the class of a group element a one means the sum formed in the group ring

$$K_a = \sum_s s^{-1} a s, \tag{1}$$

where the summation runs only over the distinct conjugates $s^{-1}as$. The classes K_a are central elements, since they commute with all group elements. They generate the center; for if an element $\sum_i \alpha_i a_i$ of the group ring commutes with all s, then

$$\sum_i \alpha_i a_i = s^{-1}\Big(\sum_i \alpha_i a_i\Big) s = \sum_i \alpha_i (s^{-1} a_i s),$$

so the coefficients are constant on conjugacy classes.

Thus the rank of the center is equal to the number of classes of conjugate group elements; hence the number of absolutely irreducible representations is also equal to this number of classes.

The matrices A and $S^{-1}AS$ have the same trace; therefore the group elements a and $s^{-1}as$ have the same trace. Taking traces on both sides of (1) yields

$$\chi(K_a) = h_a \chi(a),$$

where h_a is the number of elements in the class K_a. In words: the character of a group element is equal to the character of the class divided by the number of elements in the class.

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